

$$\Rightarrow a^2 = 1360 - 1154 = 206$$

$$\Rightarrow a \approx \sqrt{206} \approx 14$$

529. Given $1\frac{4}{5} + 20 - 280 \div 25$

$$= 1\frac{4}{5} + 20 - \frac{280}{25}$$

$$= \frac{9}{5} + 20 - \frac{56}{5}$$

$$= \frac{9 + 100 - 56}{5}$$

$$= \frac{53}{5} = 10\frac{3}{5}$$

530. $= \{(64 - 38) \times 4\} \div 13$

$$= \frac{26 \times 4}{13} = 8$$

531. Given $x + y = 2a$

$$\therefore \frac{a}{x-a} + \frac{a}{y-a} = \frac{a(y-a) + a(x-a)}{(x-a)(y-a)}$$

$$= \frac{ay - a^2 + ax - a^2}{(x-a)(y-a)}$$

$$= \frac{a(x+y) - 2a^2}{(x-a)(y-a)}$$

$$= \frac{a \cdot 2a - 2a^2}{(x-a)(y-a)}$$

$$= \frac{2a^2 - 2a^2}{(x-a)(y-a)} = 0$$

532. Let the number be x .

$$\text{Given } 421 \div 35 \times 299.97 \div 25.05 = x^2$$

$$(x)^2 \approx 420 \div 35 \times 300 \div 25$$

$$= 12 \times 300 \div 25 = 12 \times 12$$

$$\Rightarrow x^2 = 12 \times 12$$

$$\therefore x = \sqrt{12 \times 12} = 12$$

Hence, the number is 12.

533. Given

$$= 19.99 \times 15.98 + 224.98 + 125.02$$

$$\approx 20 \times 16 + 225 + 125 = 320 + 225 + 125 = 670$$

534. Given $3625 \times ? = 1450$

$$? = \frac{1450}{3625} = \frac{725 \times 2}{725 \times 5} = \frac{2}{5}$$

535. Given: $128.43 + 30.21 + ? = 173$

$$? = 173 - 128.43 - 30.21$$

$$= 173 - 158.64 = 14.36$$

536. Given expression

$$= 123 \times 999 + 123$$

$$= 123 \times (999 + 1)$$

$$= 123 \times 1000$$

$$= 123000$$

537. Given expression $\frac{(359+256)^2 + (359-256)^2}{359 \times 359 + 256 \times 256}$

$$= \frac{(359+256)^2 + (359-256)^2}{(359)^2 + (256)^2}$$

Let, $a = 359$ and $b = 256$

$$\therefore \text{expression} = \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2}$$

$$= \frac{2(a^2 + b^2)}{a^2 + b^2} = 2$$

538. $84368 + 65466 - 72009 - 13964$

$$= 149834 - 85973 = 63861$$

539. Let the number be x .

$$4376 + 3209 - 1784 + 97 = 3125 + x$$

$$\Rightarrow 7682 - 1784 = 3125 + x$$

$$\text{or, } x = 7682 - 1784 - 3125$$

$$\Rightarrow x = 2773$$

Hence, the number is 2773.

540. Let the missing number is a .

$$\text{Given } 14 \times 627 \div \sqrt{(1089)} = (a)^3 + 141$$

$$\frac{14 \times 627}{\sqrt{1089}} = (a)^3 + 141$$

$$\frac{14 \times 627}{33} = a^3 + 141$$

$$266 = a^3 + 141$$

$$a^3 = 266 - 141$$

$$a^3 = 125$$

$$a = \sqrt[3]{125} = 5$$

541. Given $\left(x + \frac{1}{x}\right) = 3$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = 3^2$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 7$$

542. As we know

$$a^3 + b^3 + c^3 - 3abc = (a^2 + b^2 + c^2 - ab - bc - ca)(a + b + c)$$

When $a + b + c = 0$

Then $a^3 + b^3 + c^3 - 3abc = 0$

When $x + y + z = 0$

$$\Rightarrow x^3 + y^3 + z^3 = 3xyz$$

$$\Rightarrow x^3 + y^3 + z^3 + 3xyz = 3xyz + 3xyz = 6xyz$$